

Q1) Complete the table with equivalents in Binary, Octal and hexa for: 143.75 (dec), 11011.101 (binary), 453 (octal), 1F.C (hex)

Sol.	Decimal	Binary	Octal	Hexa
	143.75	10001111.11	217.6	8F.C
	27.625	11011.101	33.5	1B.A
	299	100101011	453	12B
	31.75	11111.11	37.6	1F.C

Q2) Find equivalents:

a) $324_5 = ?_{10}$

Solⁿ $3 \times 5^2 + 2 \times 5^1 + 4 \times 5^0$
 $75 + 10 + 4$
 89_{10}

b) $32.23_4 = ?_{10}$

Solⁿ $3 \times 4^1 + 2 \times 4^0 + 2 \times 4^{-1} + 3 \times 4^{-2}$
 $12 + 2 + 0.5 + 0.1875$
 14.6875_{10}

$$c) 231_{10} = ?_7$$

$$\begin{array}{r|l} 7 & 231 \\ 7 & 33 - 0 \\ 7 & 4 - 5 \\ & 0 - 4 \end{array}$$

$$\therefore \underline{\underline{450_7}}$$

$$d) 189_{10} = ?_6$$

$$\text{Sol}^n \quad 189 \xrightarrow{\div}$$

$$\begin{array}{r|l} 6 & 189 \\ 6 & 31 - 3 \\ & 5 - 1 \end{array}$$

$$\therefore \underline{\underline{513_6}}$$

3) Find the Radix n :

$$a) (312/20)_n = (13 \cdot 1)_n$$

Solⁿ

$$312_n = 3n^2 + n + 2$$

$$20_n = 2n$$

$$13. \frac{1}{n} = n + 3 + \frac{1}{n}$$

$$\frac{3n^2 + n + 2}{2n} = n + 3 + \frac{1}{n}$$

$\times 2n$ on R.H.S

$$3n^2 + n + 2 = 2n^2 + 6n + 2$$

$$n(n-5) = 0$$

$$n = 5$$

$$b) 264n + 136n = 433n$$

$$\text{Sol}^n \quad 264n = 2n^2 + 6n + 4$$

$$136n = n^2 + 3n + 6$$

$$433n = 4n^2 + 3n + 3$$

$\therefore$

$$n^2 - 6n - 7 = 0$$

$$(n-7)(n+1) = 0$$

$$\therefore n = \underline{7}$$

$$c) 221505_8 = 12345_n$$

$$\text{Sol}^n \quad 2 \times 8^5 + 2 \times 8^4 + 1 \times 8^3 + 5 \times 8^2 + 5$$

$$\Rightarrow 74565_{10}$$

$$12345_7 = x^4 + 2x^3 + 3x^2 + 4x + 5$$

$x \rightarrow$ not satisfying

no x exists

4) Add following

$$\begin{array}{r} \text{a) } 00001101_2 \\ 11011111_2 \\ \hline 11101100 \end{array}$$

$$\Rightarrow \underline{11101100}_2$$

$$\text{b) } 77652$$

$$76543$$

$$\hline 176415$$

$$\Rightarrow \underline{176415}_8$$

$$\text{c) } 578A$$

$$ABDE$$

$$\hline 10368$$

$$\Rightarrow \underline{10368}_{16}$$

$$\begin{array}{r} 234 \\ 315 \\ \hline 553 \end{array}$$

$$\Rightarrow 553_6$$

→ Binary → Gray

a) 10101_2

Solⁿ Binary: $1\ 0\ 1\ 0\ 1$

Gray: $\underline{11111_2}$

b) 01110_2

Solⁿ Binary: $0\ 1\ 1\ 1\ 0$

Gray: $\underline{01001}$

→ Gray → Binary

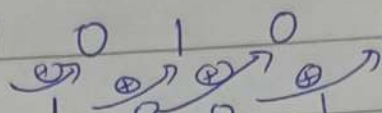
a) 01101 (Gray)

Solⁿ Gray: $0\ 1\ 1\ 0\ 1$

Binary: 01001_2

b) 10101 (Gray)

Soln Gray: $1 \ 0 \ 1 \ 0 \ 1$



Binary: 11001_2